CSC4197

RESEARCH METHODOLOGY

RESEARCH QUESTIONS & HYPOTHESES

2 FROM GAP TO QUESTION

- ✓ `Problem statement → Verified gap → Research Question → Hypothesis`
- ✓ The **RQ** is the single most-quoted sentence of your paper — reviewers judge everything against it
- ✓ A hypothesis converts a question into a **testable prediction**

3 PROPERTIES OF A GOOD RESEARCH QUESTION

A good RQ is:

- **Clear** — no vague terms ("better", "efficient" → define how measured)
- **Focused** — one question, one investigation
- **Answerable** — with data/methods you can actually obtain in your timeframe
- **Significant** — answer matters beyond your group
- **Open** — not already answered; not yes/no-trivial

Smell test: if your RQ can be answered by a single Google search, it's not a research question.

4 TYPES OF RESEARCH QUESTIONS

Type	Template	Example
Descriptive	What are the characteristics of X in context C?	What error types do first-year students make in C programming labs?
Comparative	How does A compare to B on metric M?	How does YOLOv9 compare to RT-DETR on Dhaka traffic footage (mAP, FPS)?
Relational	Is X associated with Y?	Is code-review participation associated with fewer post-release defects?
Causal	Does X cause a change in Y?	Does LLM-generated feedback improve lab scores vs. pass/fail verdicts?

5 THEORY: HYPOTHESES

- **Hypothesis** = a precise, falsifiable prediction about the relationship between variables
- **Null hypothesis (H_0):** no effect/difference exists — what statistics tries to reject
- **Alternative hypothesis (H_1):** the effect you predict
- **Directional:** "A will be *higher* than B" — use when literature predicts direction
- **Non-directional:** "A and B will differ" — when direction is unknown
- **Example:**
  - H_0: Mean lab score (LLM feedback) = Mean lab score (pass/fail only)
  - H_1: Mean lab score (LLM feedback) > Mean lab score (pass/fail only)

6 VARIABLES: THE VOCABULARY OF HYPOTHESES

- **Independent variable (IV)** — what you manipulate (feedback type)
- **Dependent variable (DV)** — what you measure (lab score)
- **Confounding variable** — sneaks in and distorts (prior CGPA, section instructor)
- **Control variables** — held constant or measured to rule out confounds

⚠️ Confounds are the #1 attack vector in your viva: *"How do you know it wasn't just the stronger students choosing the LLM group?"*

7 CONFOUNDING VARIABLE

A confound makes it difficult to determine whether the observed effect is truly due to the independent variable.

Examples:

- Students with higher previous CGPA may naturally perform better.
- One section is taught by a more experienced instructor.
- Some students already know the programming language.
- Students who use LLMs may also spend more time studying.

Suppose students choose whether to use ChatGPT.

If:

- stronger students are more likely to use ChatGPT, and
- stronger students usually earn higher marks anyway,

then higher lab scores may be due to **student ability**, not the LLM.

This is a **confounding effect**.

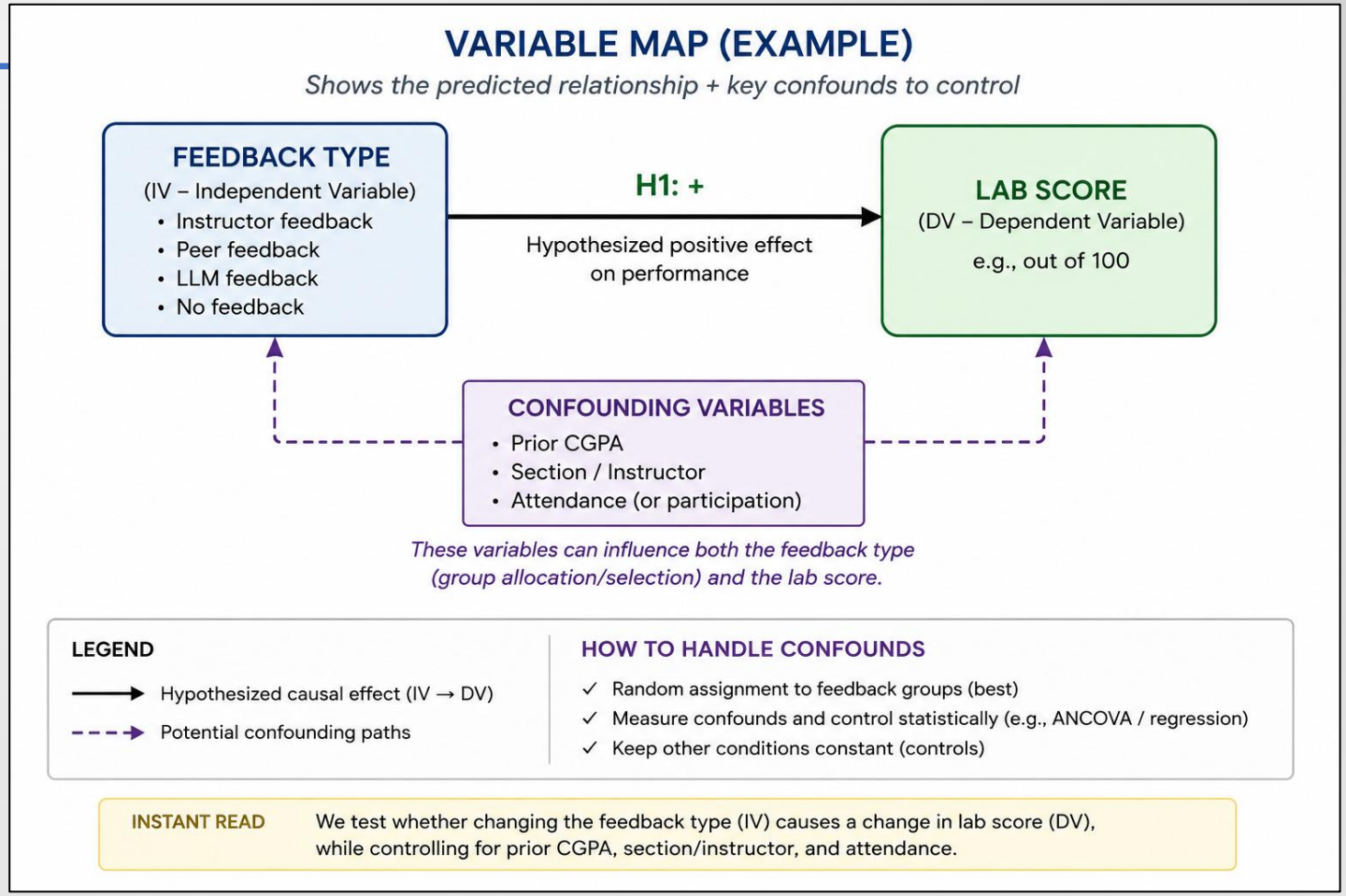
8 OPERATIONALIZATION: MAKING CONCEPTS MEASURABLE

Fuzzy concept	Operationalized measure
"Performance"	Accuracy, F1, AUROC, latency (ms), throughput (req/s)
"Usability"	SUS score (0–100), task completion time, error count
"Energy efficiency"	Joules per inference (measured via RAPL/powermetrics)
"Student engagement"	LMS logins/week, lab attendance %, time-on-task

Rule: every noun in your RQ needs a measurement plan. **"Better" is banned** unless you define the metric.

9 CONCEPTUAL FRAMEWORK DIAGRAM

- Draw the variable map before designing the study:
- One diagram = instant clarity for supervisors and reviewers
- Hands-on later: each group sketches theirs on the whiteboard



10 CASE STUDY: REFINING A WEAK RQ

- **Draft RQ (weak):** "Is AI good for education in Bangladesh?"
- **Problems:** undefined "AI", "good", "education"; population huge; not answerable in a semester.
- **Refinement steps:**
  I. Scope AI → "LLM-based programming feedback"
  II. Define "good" → "improvement in lab assessment scores"
  III. Scope population → "first-year CSE students at one private university"
- **Final RQ:** "Does LLM-generated formative feedback improve first-year CSE students' programming lab scores compared with pass/fail autograder verdicts?"
- **H_1:** Students receiving LLM feedback achieve significantly higher mean lab scores.

11 RQS IN PUBLISHED PAPERS (PATTERN RECOGNITION)

Many CS papers' structure contributions as RQ1–RQ3:

1. **RQ1 (descriptive):** What categories of X exist? → taxonomy
2. **RQ2 (comparative):** How do existing methods perform on benchmark B? → evaluation
3. **RQ3 (causal/design):** Does our proposed method improve on the best baseline? → contribution

In class Activity: Do the following tasks for your selected topic and problem:

- **Draft 3 RQs** (mix of types) — 10 min
- For the strongest RQ: write **H_0 and H_1**, identify **IV, DV, 2 confounds** — 10 min
- Sketch the **conceptual framework diagram** — 5 min